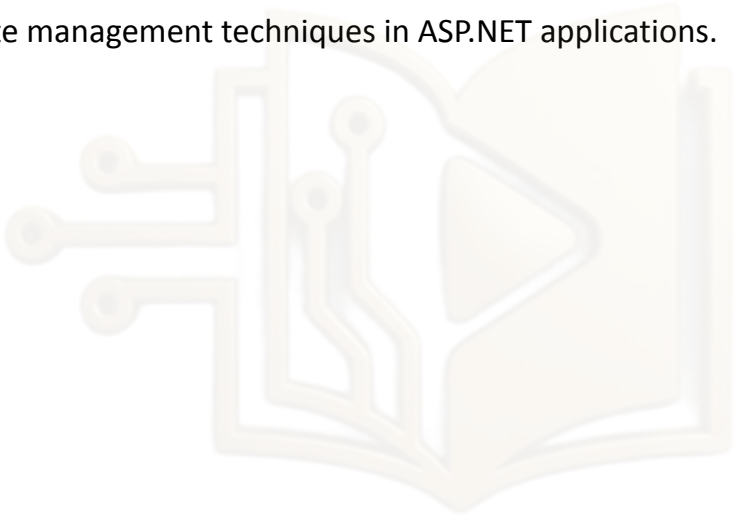
.net Technologies

1. Explain the architecture of the .NET platform in detail.
2. Compare .NET Framework, .NET Core and the unified .NET platforms in terms of runtime, tooling and application models.
3. Describe the principles of Just-In-Time (JIT) compilation. Enumerate and distinguish among its types.
4. What is a Web API? What are the advantages of using Web API?
5. Distinguish between abstract classes and interfaces in .NET. Provide use cases where one is preferred over the other.
6. Explain the middleware pipeline architecture in ASP.NET Core. How does it enhance modular development?
7. Explain the ASP.NET page life cycle. Detail the key events during its execution.
8. Describe the authentication and authorization mechanisms in ASP.NET.
9. What is a connection string in .NET? List and describe the key components of a typical SQL Server connection string
10. Discuss the concept and benefits of dependency injection in .NET Core with an implementation example.
11. Explain the role and structure of appsettings.json in ASP.NET Core projects. How does it support configuration management?
12. How can you perform CRUD operations using Entity Framework? Provide code examples for each operation.
13. Discuss the role and advantages of Entity Framework Core in data access operations.
14. Describe structured exception handling in C# using try, catch, and finally constructs.

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15. What is Language Integrated Query (LINQ)? Explain how LINQ enhances data querying within .NET applications.
16. Discuss asynchronous programming in .NET with reference to async and await keywords. How do they improve application responsiveness?
17. Discuss the benefits and challenges of adopting a microservices approach in enterprise .NET applications.
18. Discuss the Repository design pattern. Explain how it abstracts data access logic and promotes separation of concerns in ASP.NET applications.
19. Discuss the benefits of applying SOLID principles in software development.
20. Describe the Garbage Collection mechanism in .NET.
21. Explain state management techniques in ASP.NET applications.



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